

VRUDDHI MULE

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Professional Summary

Artificial Intelligence and Data Science undergraduate with Honors in Data Engineering and internship experience in machine learning, analytics, forecasting, and recommendation systems. Experienced in building AI applications, ETL pipelines, RESTful APIs, and geospatial machine-learning workflows using Python, SQL, FastAPI, PyTorch, and PostgreSQL. Current research includes combining monitoring, satellite, and weather data for urban air-quality mapping with gradient boosting and U-Net models.

Education

KJ Somaiya School of Engineering
Bachelor of Technology in Artificial Intelligence and Data Science
Honors in Data Engineering

Mumbai, India
Expected 2027

Experience

KEC International
Data Analytics and Machine Learning Intern

Mumbai, India
May 2025 – Aug 2025

- Built an end-to-end sales forecasting pipeline covering preprocessing, transformation, validation, feature engineering, scaling, and model-ready dataset creation.
- Performed exploratory data analysis to identify trends, anomalies, correlations, and business drivers affecting sales performance.
- Developed dashboards and visualizations to translate analytical findings into actionable insights for business stakeholders.

KJ Somaiya College of Engineering
Artificial Intelligence Intern

Mumbai, India
Dec 2024 – Apr 2025

- Developed an AI-powered recommendation engine using a fine-tuned BERT model and semantic similarity, contributing to a 30% increase in student participation.
- Automated web scraping, collection, preprocessing, and labeling of 5,000+ hackathon records to create structured, model-ready datasets.
- Built an NLP recommendation pipeline using text embeddings and cosine similarity and integrated it with internal university platforms.

Projects

Aitken – Geospatial PM_{2.5} Mapping

Ongoing

- Developed a retrospective daily air-quality mapping pipeline for Greater London at 1 km resolution, preparing 444,202 station-day observations from LAQN and Breathe London across 2021–2024.
- Integrated monitoring records, ERA5 weather, Sentinel satellite products, and geographic covariates; engineered 66 predictors, including strictly prior-day PM_{2.5} history, across 1,719 grid cells.
- Implemented and compared histogram gradient boosting, XGBoost, ANN, hybrid HGB + residual U-Net, and standalone U-Net models using Scikit-learn and PyTorch, including ensembles across multiple random seeds.
- Achieved RMSE of 3.78 $\mu\text{g}/\text{m}^3$, MAE of 2.34 $\mu\text{g}/\text{m}^3$, and $R^2 = 0.518$ with the hybrid ensemble on a retrospective 2024 benchmark of 9,619 regulatory station-days; produced spatial maps and audited saved predictions.

CareerPilot AI – AI-Powered Career Intelligence Platform

Jul 2026

- Built and deployed an AI platform using Next.js, TypeScript, FastAPI, and Google Gemini API for resume analysis and intelligent job recommendations.
- Developed a resume-processing and ATS analysis pipeline to extract candidate information, identify skill gaps, evaluate resume quality, and recommend suitable career roles.
- Integrated LLMs to rank job opportunities against candidate profiles and generate match scores, matched and missing skills, learning priorities, and interview-readiness insights.
- Designed RESTful APIs using FastAPI and Pydantic, integrated Clerk authentication, and deployed the frontend on Vercel and backend on Render.

AI Promotion and Price Optimizer

Jan 2026

- Developed a dynamic pricing application using FastAPI, XGBoost, and Next.js to generate data-driven e-commerce pricing recommendations.
- Engineered a market-intelligence pipeline using SerpApi and custom web crawlers to collect and process competitor pricing across multiple marketplaces.
- Implemented feature-engineering workflows to prepare pricing, inventory, and market data for machine-learning models.
- Designed backend workflows using Supabase and Prisma ORM and integrated Slack and n8n for human-in-the-loop approval of pricing recommendations.

FloatChat – LLM-Powered Data Analytics Platform

Sep 2025

- Built a conversational analytics platform enabling natural-language querying and visualization of ARGO oceanographic datasets.
- Developed an ETL pipeline to ingest and transform NetCDF datasets into PostgreSQL and Chroma vector databases for structured and semantic retrieval.
- Implemented a Model Context Protocol (MCP) backend to translate natural-language requests into executable database queries.
- Developed interactive geospatial dashboards using Plotly and Leaflet to explore and visualize processed oceanographic data.

Technical Skills

Programming Languages: Python, SQL, C++, JavaScript, TypeScript

Machine Learning and AI: Scikit-learn, PyTorch, XGBoost, Histogram Gradient Boosting, U-Net, Neural Networks, BERT, NLP, LLMs, Generative AI, Google Gemini API, Text Embeddings, Semantic Search, Recommendation Systems, Feature Engineering, Model Evaluation, Time-Series Forecasting

Data Science and Geospatial Analytics: Pandas, NumPy, EDA, Statistical Analysis, Data Cleaning, Preprocessing, Transformation, Visualization, GeoPandas, Rasterio, Xarray, Satellite Data Processing

Data Engineering and Databases: PostgreSQL, ETL/ELT Pipelines, Data Ingestion, Data Validation, Data Quality, Relational Databases, Vector Databases, ChromaDB, Supabase, Web Scraping

Backend and Software Development: FastAPI, Pydantic, RESTful APIs, Node.js, React.js, Next.js, API Integration, Object-Oriented Programming

Tools and Platforms: Git, GitHub, Linux, Postman, Vercel, Render, Plotly, Leaflet

Familiarity / Coursework: Apache Spark, PySpark, Apache Kafka, Apache Airflow, AWS, Snowflake, Databricks, dbt, Power BI

Computer Science Fundamentals: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Database Design, API Design

Achievements

- Selected among the top 100 participants nationwide in the ZS Campus Beat Challenge 2026.
- Participated in national-level hackathons, developing AI, data engineering, analytics, and full-stack solutions under time constraints.

Leadership and Volunteering

Marketing Team Member, Somaiya Space Research Program – Coordinated outreach and marketing initiatives to increase student engagement in space-research programs and technical events.